HRS Group Platform Engineer Technical Assessment

Thank you for your interest in joining HRS Group! As part of our selection process, all of our candidates must take the following test.

We think platform engineering is about building robust, scalable infrastructure that enables engineering teams to move fast while maintaining reliability and observability.

We are testing your ability to design and implement modern cloud-native platforms, as well as your understanding of multi-tenancy, observability, and platform automation. In your solution you should emphasize scalability, maintainability, and platform engineering best practices.

To begin, create a GitHub repository and start adding your work. Commit often, we would rather see a history of trial and error than a single monolithic push. When you're finished, send us the URL to your repository.

You can use the following folder structure or create your own:

```
./
├─ 1_platform_design
│   └─ <your project>
├─ 2_infrastructure
│   └─ <your project>
└─ 3_observability
    └─ <your project>
```

Duration: 2 to 4 hours

1. Platform Design Test

Scenario: Design and build a multi-tenant Application platform with CI/CD that supports 20+ engineering teams (250+ engineers) with the ability to scale to 50+ teams.

Requirements:

1. Architecture Document:

- High-level architecture diagram (PNG/SVG format)
- Multi-tenancy isolation strategy and security model
- Scalability approach (20 → 50+ teams)
- Cost estimation for AWS eu-central-1 region
- Key design trade-offs and rationale

2. Infrastructure as Code:

- Terraform or CloudFormation implementation:
  - VPC with container orchestration (EKS/ECS)
  - Multi-tenant namespace configuration with proper isolation
  - Sample CI/CD pipeline infrastructure
  - Storage solutions (RDS, S3/artifact storage)
  - RBAC and IAM roles
  - Network policies and security groups

3. Documentation:

- Setup and deployment instructions
- Design decisions justification
- Security best practices implementation

Bonus Points:

- Detailed multi-tenancy isolation mechanisms
- Production-ready security implementation
- Cost optimization strategies
- Comprehensive architecture diagrams
- Disaster recovery considerations

2. Observability Test

Implement observability for your platform using OpenTelemetry and New Relic. Build a comprehensive monitoring solution that provides insights into pipeline performance, resource utilization, and tenant-specific metrics. Export data should be structured for analysis and alerting.

Bonus points:

- OTLP exporter configuration with proper data collection.
- Sample dashboards showing key platform metrics (SLIs/SLOs).
- Tenant isolation in monitoring data.
- Documentation of monitoring strategy and key performance indicators.